EVANS KWASI

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RESEARCH INTERESTS

Human-Centered AI; Speech and Language Technologies; Low-Resource Language Processing; Multimodal Large Language Models; Adaptive AI and User Modelling

EDUCATION

M.Sc. Computer Science | *University of Ghana, Accra* Sep 2024 – Mar 2026

- Thesis: Towards the Development of a Maternal Health Chatbot: A Single-Speaker Model
- CGPA: 3.62 / 4.0

B.Sc. Computer Science – First Class | *KNUST, Kumasi* Sep 2018 – Nov 2022

- CWA: 72.3 / 100 | Project: Deep Learning & IoT for Precision Agriculture — Automated Weed Detection and Multi-Sensor Soil Monitoring

RESEARCH/PROFESSIONAL EXPERIENCE

Researcher / Developer | *Interactive Systems Research Group, York University* Nov 2025 – Present

- Investigate adaptive mnemonic generation using multimodal large language models and genetic algorithms to support memorability and long-term recall.
- Led a systematic review on persuasive technology in the circular economy as corresponding author; manuscript prepared for submission.
- Led testing of recall from mnemonic approaches with 20 undergraduate participants.
- Lead an empirical study on cross-device interface consistency as corresponding author; manuscript in preparation.

Research Assistant & AI Engineer | *HCI Lab, University of Ghana* Jun 2023 – Present

- Contribute to speech technology development for low-resource African languages, including Asante-Twi and Fante.
- Lead the speech synthesis development team in collaboration with Google Ghana, from data collection to model development and evaluation.
- Fine-tuned and pretrained various speech synthesis models on Asante-Twi and Fante.
- Fine-tuned multilingual conversational models on culturally grounded health data.
- Contribute to automatic speech recognition modelling for standard and speech-impaired speakers.
- Led the collection and development of a multilingual maternal health dataset (Twi, Fante, Ga, Ewe, Baoule)
- Develop speech and conversational AI components for culturally grounded health chatbot, integrating the retrieval-augmented generation, machine translation and speech synthesis pipeline.
- Developed and managed lab's website.

Lead Data Scientist | *St. Michael's Hospital Research Lab, Jachie Pramso* Nov 2022 – May 2024

- Developed an explainable ML model for diabetes prediction (RF, SVM, XGBoost, ANN) trained on a custom Ghanaian clinical dataset; achieved 99.3% accuracy across classical models.
- Applied SHAP (SHapley Additive exPlanations) for both local and global model explainability—waterfall plots, mean SHAP feature importance, and dependence plots—enabling clinician-facing transparency.
- Conducted statistical validation using accuracy, precision, recall, F1-score, and ROC-AUC; contributed to academic publication of findings.

Software & AI Engineer(Co-founder) | *Techline Systems Ghana,Takoradi* Sep 2021 – Present

- Build software solutions to support inventory tracking, sales management, and day-to-day business operations for small businesses in rural communities.
- Deploy CCTV camera systems for small shops and mini-marts, supporting business security and monitoring needs.
- Develop analytics dashboards that help business owners monitor sales, review business performance, and make data-informed operational decisions.

TEACHING/PROFESSIONAL EXPERIENCE

Teaching Assistant, Department of Computer Science, KNUST Nov 2022 – May 2024

- Taught Introduction to Computing to 500+ undergraduate students in Economics, Geography, and Veterinary Medicine.
- Organized and facilitated programming tutorials to reinforce programming concepts and practical skills.
- Held office hours to provide academic support and clarify course material.
- Graded coursework and examinations to assess student learning and performance.
- Mentored 50 final-year undergraduate students on their academic projects.

PUBLICATIONS

- Mensah, M. A., **Kwasi, E.**, Abdulmawjood, M., & Oyibo, K. (2026, June). User Modelling and Adaptation in Immersive Preventive Health Interventions: A Systematic Review and Evidence Map. In *Proceedings of the 34th ACM Conference on User Modeling, Adaptation and Personalization* (pp. 263-272).
- Wiafe, I., Ekpezu, A. O., Winful, F. B. P., Wiafe-Akenten, A. N., Mensah, M. A., Semanyoh, N., Salihs, S. A., **Kwasi, E.**, Apatti, J. K., Odame, G., Osei, K. O., & Yeboah, F. E. (2026). A Parallel English–Akan Maternal Health Dataset to Support Machine Translation and Text-to-Speech Systems. *Data in Brief*. DOI:10.1016/j.dib.2026.113088
- Mensah, M. A., Oyibo, K., Rahman, P., **Kwasi, E.**, & Adaji, I. (2026). From plastic waste awareness to circular action: A systematic review of persuasive technologies for sustainable plastic circularity. *Sustainable Development.* (under review)

CONFERENCE PRESENTATIONS

- **Kwasi, E**., Wiafe, I., et al. (2025). Developing an Akan Text-to-Speech System: Corpus Creation, Neural Modeling, and Digital Inclusion in Ghana. Presented at SOLCON V, University of Ghana.
- Bekor, C., Wiafe, I., & **Kwasi, E**. (2025). AI-Powered Image Description in Akan and Ewe. Presented at SOLCON V, University of Ghana.
- **Kwasi, E.**, Rose-Mary Gyening. Diabetes classification Using Machine Learning Techniques and Explainable Artificial Intelligence. Poster presentation at the Deep Learning IndabaX Ghana Conference, Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana, 2024.
- Wiafe, I., Semanyo, N. K., **Kwasi, E**., Ekpezu, A., & Winful, P. (2025). *Advancing Inclusive, Safer and Ethical Digital Space: Piloting the Use of AI to Detect Hate and Offensive Speech in Low-Resource Languages in Ghana (Akan). Presented at SOLCON V, University of Ghana, Legon, Accra, Ghana, 2025.

LEADERSHIP, VOLUNTEERING & COMMUNITY SERVICE

Mentor, Scholarship and Academic Guidance Initiative(bizame.app) 2021 – Present

- Mentor senior high school graduates through scholarship applications and provide guidance on selecting university programmes aligned with their interests and career aspirations.
- Supported 4 students in securing fully funded scholarships.

Hackathon Mentor, MTN Ghana Tɛkyɛrema Pa Hackathon (youtube link) Sep – Oct 2026

- Mentored 25 teams developing technology solutions, providing guidance to strengthen their ideas and approaches.

Organizer and Host, Tɛkyɛrema Pa Hackathon Apr 2025 – May 2025

- Supported hackathon organization and hosted 25 participants on campus, facilitating participation in the event.

President & Senator, Computer Science Department, KNUST 2020 – 2022

- Led student representation and supported initiatives to advance student interests within the Computer Science Department.
- Founded a skills-training club to tackle high programming course trails and development of market ready skills.
- Lobbied for increment of student data bundles from 2GB to 4GB.

Teaching, Esiama 2020 – 2023

- Volunteered to provide academic support to 3 public junior high schools.
- Volunteered to teach Grade 1 pupils for one academic term, supporting learning and classroom instruction.

Class President, KNUST 2018 – 2022

- Coordinated with teaching assistants (TAs) to organize tutorials in courses students found challenging.
- Advocated for flexible deadlines and extended access to online learning resources and examinations during COVID-19 for students in rural areas
  Advocated for reasonable accommodations for students with disabilities

AWARDS

- Mastercard Foundation Scholars Program, KNUST 2018 - 2022

REFERENCES

Prof. Isaac Wiafe | University of Ghana | iwiafe@ug.edu.gh; Assoc. Prof. Kiemute Oyibo | York University |

kiemute.oyibo@yorku.ca Dr.